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## 1.0 Introduction

### 1.1 Data Description

The dataset worked in this study contains **211,913 logs (rows) and 12 variables (columns)**, recording web hacking incidents over the past 15 years. It involves characteristics such as the date of destruction, the reporting entity, the compromised website URL, and the associated IP address. Furthermore, it gives information on the affected server's place, operating system, and web server version. The dataset also reports determining details, the major language of the defaced website, and financial effects such as ransom paid, system downtime in days, and revenue lost due to the attack. This dataset serves as the foundation for analyzing hacking trends, identifying vulnerabilities, and deriving actionable insights to enhance cybersecurity measures.

### 1.2 Assumptions

This analysis focusses on the dataset provided in which is comprehensive and precisely indicates web hacking incidents over the past 15 years. It is assumed that each recorded web hacking incident is appropriately linked to the corresponding country and year, ensuring reliability in trend analysis. This analysis emphasizes that all data pre-processing, modifications, and visualization steps are executed using R programming without third-party tools, as per assignment requirements. On top of that, data anomalies, such as major spikes in web hacking incidents for specific years or countries, are either legitimate incidents or require additional evidence before analysis. Subsequently, this analysis signifies that cyber threats are constantly emerging, which leads to trends discovered in the dataset may not completely indicate existing or potential cyber security concerns, highlighting the importance of extensive research and analysis.

## **1.3 Hypothesis and Objectives**

### **Hypothesis**

"The frequency and severity of web hacking incidents have increased over the past 15 years, with a significant correlation between notifiers, targeted platforms, and the geographical distribution of attackers, contributing to higher financial losses and downtime for organizations."

### **Objectives**

1. To analyze the frequency and patterns of web attacks to identify trends over the years.
2. To investigate the relationship between financial losses and downtime due to web hacking incidents over the years.
3. To investigate relationships between web hacking incidents and targeted platforms.
4. To Investigate the relationship between key geographical trends and yearly trend comparison of web hacking incidents.

## 2.0 Data Preparation

### 2.1 Data Import

```
#IMPORT
#Read the File
# Load the dataset
data <- read_csv("C:/Users/Adam JJ/Desktop/SEM 1/PFDA/hackingData.csv", locale = locale(encoding = "latin1"))

#INSTALL LIBRARY
#install.packages("dplyr")
#install.packages("readr")
#install.packages("stringr")

# Load necessary packages
library(dplyr)
library(readr)
library(stringr)
```

*Figure 1: Loading dataset into R studio*

Load the dataset in R studio and install the necessary packages needed to clean the data.

### 2.2 Data Cleaning

```
#DATA CLEANING

# Ensure column names are clean
colnames(data) <- make.names(colnames(data), unique = TRUE)
```

*Figure 2: Cleaning columns*

```
# Trim whitespace, remove duplicates
data <- data %>%
  mutate(across(where(is.character), str_trim)) %>%
  distinct()
```

*Figure 3: Removing duplicates / whitespaces*

```
# Convert numeric columns and filter invalid values
data <- data %>%
  mutate(
    Ransom = as.numeric(Ransom),
    Loss = as.numeric(Loss),
    DownTime = as.numeric(DownTime)
  ) %>%
  filter(if_else(!is.na(DownTime), DownTime >= 0, TRUE)) %>%
  filter(if_else(!is.na(Loss), Loss >= 0, TRUE)) %>%
  filter(if_else(!is.na(Ransom), Ransom >= 0, TRUE))
```

*Figure 4: Filtering invalid values*

```
# Standardize country names and remove "Unknown" as NA
data <- data %>%
  mutate(
    Country = str_trim(Country),
    Country = str_to_title(Country),
    Country = case_when(
      Country == "United State" ~ "United States",
      Country == "United Kingd" ~ "United Kingdom",
      Country == "Viet Nam" ~ "Vietnam",
      Country == "Unknown" ~ NA_character_,
      TRUE ~ Country
    )
  )
```

*Figure 5: Standardizing countries and unknown values*

A few examples of how we unified certain Countries which had slightly different spelling or unnecessary symbols.

## 2.3 Data Validation

```
#DATA VALIDATION
#DATASET INFORMATION
str(data)
```

*Figure 6: Dataset information*

```
#SUMMARY OF ENTIRE DATA
summary(data)
```

*Figure 7: Summary of data*

```
#SUMMARY OF A PARTICULAR COLUMN
summary(data$Date)
summary(data$Notify)
summary(data$URL)
summary(data$IP)
summary(data$Country)
summary(data$OS)
summary(data$WebServer)
summary(data$Encoding)
summary(data$Lang)
summary(data$Ransom)
summary(data$DownTime)
summary(data$Loss)
```

*Figure 8: Summary of each column from dataset*

```
#MAX VALUES
max(data$Ransom)
max(data$DownTime)
max(data$Loss)
```

*Figure 9: Minimum Values for Numbered Data*

```
#MIN VALUES
min(data$Ransom)
min(data$DownTime)
min(data$Loss)
```

*Figure 10: Maximum Values for Numbered Data*

```
# Debugging: Check if changes were applied before saving
print(unique(data$Country)) # See if country names updated
sum(is.na(data$Country)) # Count NA values
```

*Figure 11: Debugging for changes made*

## 3.0 Data Analysis

### 3.1 Objective 1: To analyze the frequency and patterns of web attacks to identify trends over the years. – Adam Jerome Joseph (TP071194)

#### 3.1.1 Analysis 1-1: Yearly Trend of Web Hacking Incidents

```
# Convert Date Column To Date Format  
CleanedData$Date <- as.Date(CleanedData$Date, format="%m/%d/%Y")
```

*Figure 12: Date Formatting*

```
# Extract The Year From The Date Column  
CleanedData$Year <- year(CleanedData$Date)
```

*Figure 13: Extraction of Year*

```
# Count Incidents Per Year  
yearly_trends <- CleanedData %>%  
  group_by(Year) %>%  
  summarise(Incidents = n()) %>%  
  arrange(Year)
```

*Figure 14: Incidents Per Year*

```
# Plotting of the yearly trend of hacking incidents  
ggplot(yearly_trend, aes(x = Year, y = Num_Incidents)) +  
  geom_bar(stat = "identity", fill = "steelblue") +  
  theme_minimal() +  
  labs(title = "Yearly Web Hacking Incidents Trend", x = "Year", y = "Number of Incidents") +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

*Figure 15: Plotting of yearly trends*

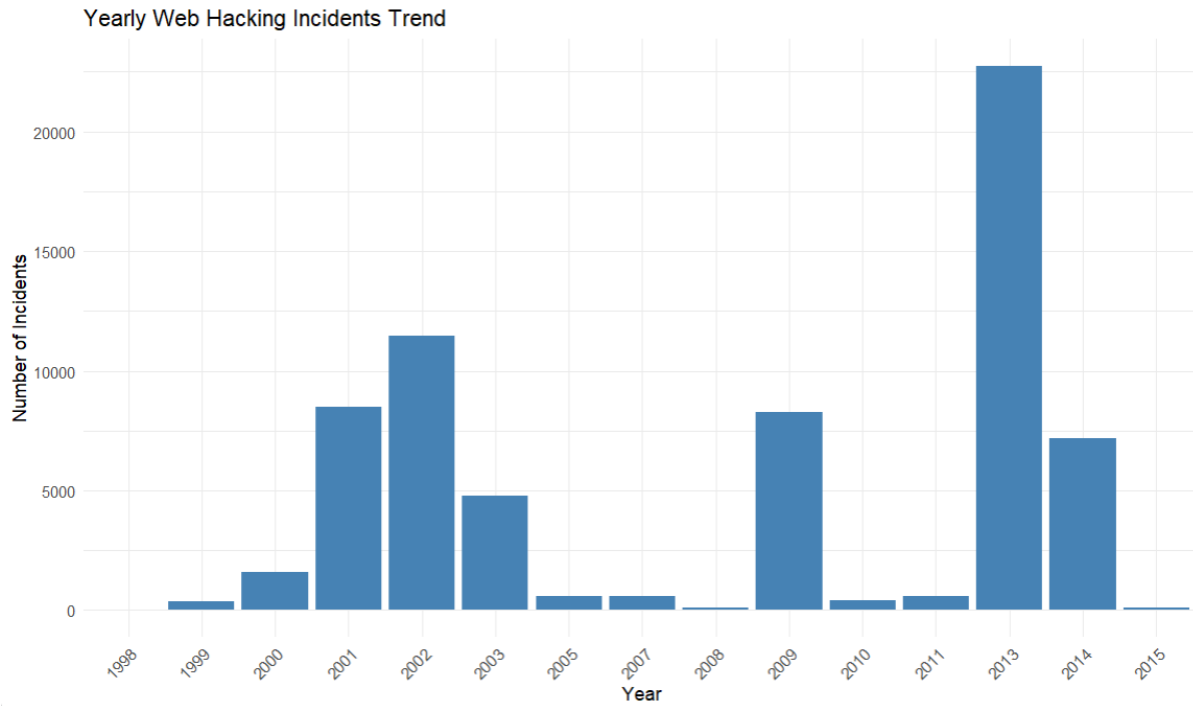


Figure 16: Bar Chart of Yearly Web Hacking Incidents

|    | Year | Incidents |
|----|------|-----------|
| 1  | 1998 | 18        |
| 2  | 1999 | 375       |
| 3  | 2000 | 1568      |
| 4  | 2001 | 8489      |
| 5  | 2002 | 11461     |
| 6  | 2003 | 4791      |
| 7  | 2005 | 582       |
| 8  | 2007 | 593       |
| 9  | 2008 | 120       |
| 10 | 2009 | 8258      |
| 11 | 2010 | 414       |
| 12 | 2011 | 573       |
| 13 | 2013 | 22749     |
| 14 | 2014 | 7183      |
| 15 | 2015 | 83        |
| 16 | NA   | 144656    |

Figure 17: Display of Incidents in each Year

The **goal** of this analysis is to investigate the **yearly frequency of web hacking incidents** over the past 15 years. **Identifying trends** in cyberattacks can help security experts, policymakers, and organizations implement effective cybersecurity strategies. A higher frequency of attacks in specific years may indicate emerging threats, vulnerabilities, or increased hacker activities.

### Graph Interpretation:

- The **x-axis** represents the **year** when the incident occurred.
- The **y-axis** represents the **number of hacking incidents** recorded for that year.
- **Bars with higher values indicate peak hacking activity**, showing an increase in web attacks.
- If the trend shows a **steady rise in incidents**, it suggests growing cyber threats, whereas a **sudden drop** can indicate improved security measures.

### Findings:

The chart shows web hacking incidents from 1998 to 2015. In the late 1990s (1998-1999), incidents were minimal due to limited internet adoption and simpler web technologies. The internet was still developing, with fewer websites and less reliance on online platforms, which resulted in fewer opportunities for hackers. However, from 2000 to 2002, incidents surged dramatically, peaking in 2002 with over 10,000 cases. This increase coincided with the dot-com boom, where rapid online business growth created a vast attack surface for cybercriminals. During this period, hackers exploited emerging vulnerabilities like SQL injection and cross-site scripting (XSS), which were poorly understood by developers at the time. Additionally, the rise of hacktivist groups targeting corporations for ideological reasons also contributed to the surge.

Following this peak, incidents declined sharply and stabilized between 2003 and 2008. This decline can be attributed to growing security awareness and the adoption of better defensive measures, such as firewalls, intrusion detection systems, and secure coding practices. Governments began introducing legislation to combat cybercrime, deterring some attackers. Companies also

prioritized cybersecurity after the massive wave of incidents in the early 2000s, which had likely highlighted the financial and reputational risks of poor security.

In 2009, incidents rose again, reflecting new challenges. The adoption of Web 2.0 technologies—characterized by dynamic, user-driven content—introduced fresh vulnerabilities. Attackers targeted interactive websites, social media platforms, and cloud services that had grown popular during this time. Furthermore, the global financial crisis of 2008 might have motivated financially driven cybercriminals, as a rise in cybercrime often coincides with economic instability.

A significant spike occurred in 2013, with over 20,000 reported incidents, marking the highest point in the chart. This surge was likely fueled by the proliferation of automated cybercrime tools like exploit kits, which made hacking easier and more accessible. The increasing number of high-profile breaches in 2013, such as those affecting major companies like Yahoo and Adobe, also underscored the vulnerabilities in web systems, leading to more incidents being reported or detected. Additionally, the rapid growth of internet-connected devices (IoT) and online services expanded the attack surface for cybercriminals.

After 2013, incidents declined but remained higher than pre-2013 levels. This decline likely reflects organizations' increased investment in advanced cybersecurity measures, such as encryption, real-time threat detection, and compliance with stricter data protection standards. Public awareness of cybersecurity risks and collaboration between governments and private companies may also have contributed to the decrease. However, the relatively high levels post-2013 highlight the ongoing challenges posed by the ever-evolving tactics of hackers, driven by advancements in technology and increasing reliance on digital platforms.

### 3.1.2 Analysis 1-2: Top Notifiers for Web Attacks

```
# Standardize notifier names (remove leading/trailing spaces, fix case inconsistencies)
data <- data %>%
  mutate(Notify = str_trim(Notify))
```

*Figure 18: Standardizing Notifier Names*

```
# Total number of incidents for each notifier
notifier_counts <- data %>%
  group_by(Notify) %>%
  summarise(Num_Reports = n(), .groups = "drop") %>%
  arrange(desc(Num_Reports))
```

*Figure 19: Finding total number of incidents for each notifier*

```
# Check the top 10 notifiers
print(head(notifier_counts, 10))
```

*Figure 20: Checking the top 10 notifiers*

```
# Select the top 10 notifiers
top_notifiers <- notifier_counts %>%
  slice_head(n = 10)
```

*Figure 21: Selecting the top 10 notifiers*

```
# Plotting top 10 notifiers using ggplot2
ggplot(top_notifiers, aes(x = reorder(Notify, -Num_Reports), y = Num_Reports)) +
  geom_bar(stat = "identity", fill = "blue") +
  coord_flip() + # Flip coordinates for better readability
  theme_minimal() +
  labs(title = "Top 10 Notifiers of Web Attacks", x = "Notifier", y = "Number of Reports")
```

*Figure 22: Plotting top 10 notifiers*

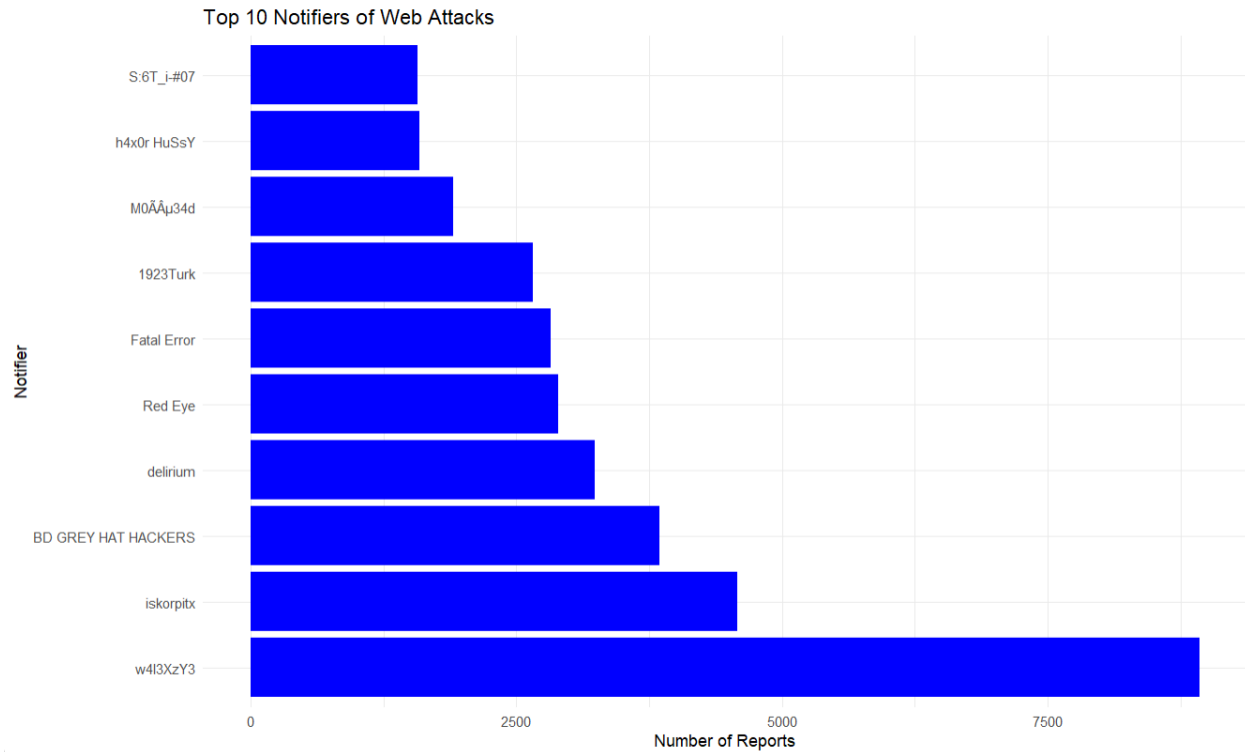


Figure 23: Top 10 Notifiers of Web Attacks

|    | Notify                | Num_Reports |
|----|-----------------------|-------------|
|    | <chr>                 | <int>       |
| 1  | "w4l3XzY3"            | 8928        |
| 2  | "iskorpitx"           | 4577        |
| 3  | "BD GREY HAT HACKERS" | 3849        |
| 4  | "delirium"            | 3238        |
| 5  | "Red Eye"             | 2891        |
| 6  | "Fatal Error"         | 2826        |
| 7  | "1923Turk"            | 2652        |
| 8  | "M0Ä\u0082Äµ34d"      | 1910        |
| 9  | "h4x0r HuSsY"         | 1589        |
| 10 | "S:6T_i-#07"          | 1569        |

Figure 24: Display of number of reports for each notifier

The goal of this analysis is to examine the **top notifiers** that report web hacking incidents. By identifying the most active notifiers, we can determine **which entities are most engaged** in cybersecurity monitoring, understand **how reporting activity correlates with attack frequency** and identify **potential gaps in reporting** maybe due to underreporting or lack of monitoring.

### Graph Interpretation:

- The **x-axis** represents the **notifiers** (entities that reported hacking incidents).
- The **y-axis** represents the **number of incidents reported** by each notifier.
- Higher bars indicate **more active notifiers**, while lower bars suggest **less engagement in attack reporting**.

### Findings:

The bar chart presents the top 10 notifiers of web attacks, highlighting the number of hacking incidents reported by each entity. These notifiers, which could be individuals, hacker groups, or automated security systems, play a crucial role in identifying and documenting web-based cyber threats. The chart shows that the most active notifier, **w4l3XzY3**, has reported the highest number of incidents, significantly more than the others. Other notable notifiers, such as **iskorpitx**, **BD GREY HAT HACKERS**, and **delirium**, have also made substantial contributions. Identifying the top notifiers is important for several reasons. First, it enhances threat intelligence by determining where most cybersecurity reports originate. If only a handful of sources contribute many reports, verifying their reliability and methodologies becomes essential. Additionally, analysing data from these notifiers helps detect patterns and trends in cyberattacks, allowing security professionals to anticipate emerging threats and develop preventive measures.

Moreover, recognizing key notifiers facilitates collaboration and verification, ensuring that the data being used to strengthen cybersecurity defences is accurate and credible. It also helps in identifying potential malicious actors who may be manipulating threat intelligence for their own advantage. Some notifiers may be providing misleading data, either unintentionally or as part of an attempt to disrupt cybersecurity efforts. Knowing which sources are most active in reporting

incidents allows organizations to prioritize credible reports and streamline their response mechanisms, ensuring quicker and more effective mitigation of cyber threats.

This information can be leveraged in several ways. Organizations and security researchers can validate and cross-check reports from the leading notifiers to filter out false positives. Encouraging broader participation in reporting can diversify the sources of cyber threat intelligence, reducing reliance on a limited set of notifiers. Security teams can also use insights from these reports to strengthen their defences against the most reported attack methods. Additionally, collaborating with top notifiers can provide a deeper understanding of emerging attack strategies and vulnerabilities. If a particular notifier is reporting an excessively high number of incidents, further investigation may be necessary to determine whether the data is genuine or artificially inflated.

### 3.2 Objective 2: To investigate the relationship between financial losses and downtime due to web hacking incidents over the years – Hemanthrajah A/L N Krishnarajah (TP070844)

```
# Load necessary libraries  
library(ggplot2)  
library(reshape2)
```

*Figure 25 Load libraries*

```
# Read the CSV file  
df <- read.csv("C:\\Users\\heman\\Downloads\\cleaned_hackingData.csv", stringsAsFactors = FALSE)
```

*Figure 26 Loading the cleaned csv file*

```
# Convert the 'Date' column to Date type and extract year (assuming format yyyy-mm-dd)  
df$Date <- as.Date(df$Date, format = "%Y-%m-%d")  
df$year <- as.numeric(format(df$Date, "%Y"))  
df$month <- as.numeric(format(df$Date, "%m"))
```

*Figure 27 Covert data type*

```
# Remove rows with NA values in critical columns  
df <- df[!is.na(df$Loss) & !is.na(df$year) & !is.na(df$month), ]
```

*Figure 28 Remove rows for better interpretation*

### 3.2.1 Analysis 1-1: The relationship between Year, Downtime, and Loss

```
#-----  
# Analysis 1. MATRIX PLOT: The relationships between Year, Downtime, and Loss  
#-----  
# Create a subset with only the numeric columns needed  
matrix_data <- data.frame(  
  Year = df$year,  
  Downtime = df$Downtime,  
  Loss = df$Loss  
)
```

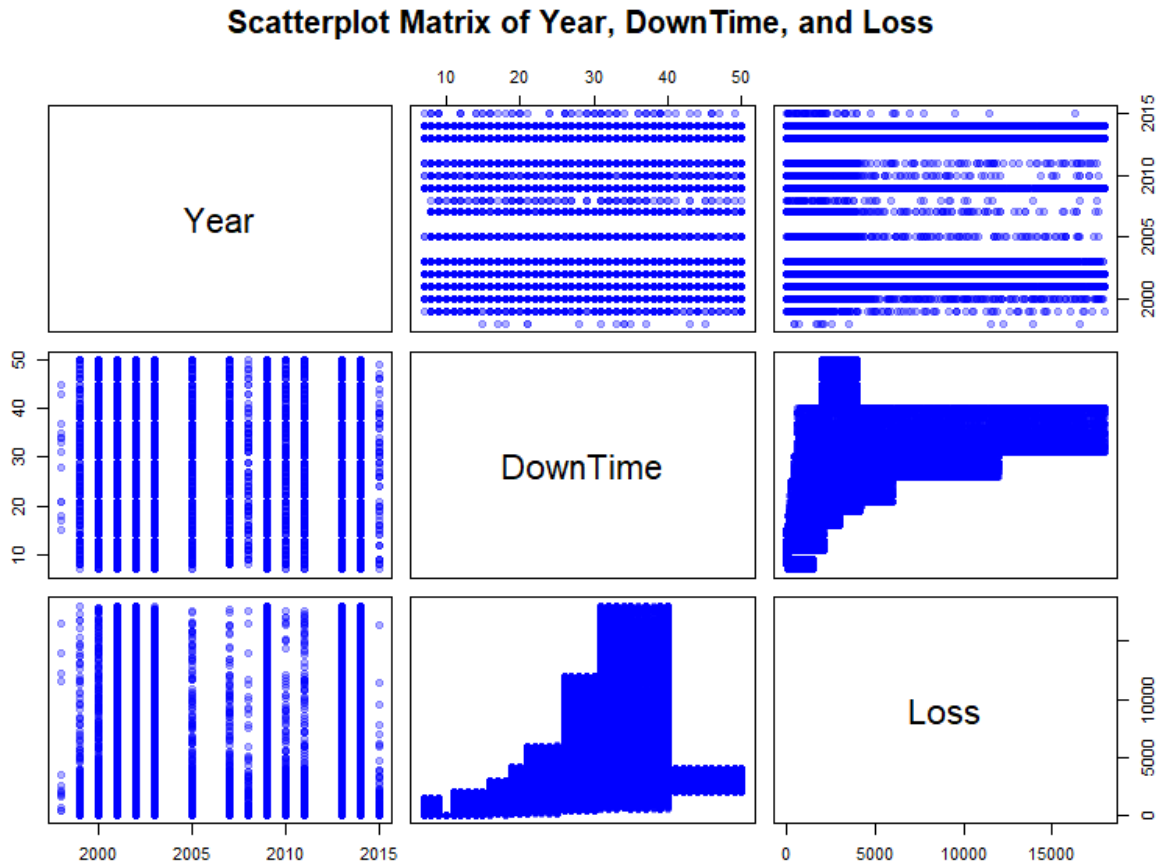
*Figure 29 Creating subset*

```
# Save scatterplot matrix as an image  
png("matrix_plot_final.png", width = 800, height = 600, res = 100)  
pairs(matrix_data,  
  main = "Scatterplot Matrix of Year, Downtime, and Loss",  
  pch = 19,  
  col = rgb(0, 0, 1, 0.3),  
  labels = c("Year", "Downtime", "Loss"))  
dev.off()
```

*Figure 30 Saving the scatterplot*

```
#-----  
# Print Summary Statistics and Correlation Matrix  
#-----  
cat("\nSummary Statistics:\n")  
print(summary(matrix_data))  
  
cat("\nCorrelation Matrix:\n")  
print(cor(matrix_data, use = "complete.obs"))
```

*Figure 31 Print Summary Statistics*



*Figure 32 Scatterplot Matrix of Year, DownTime, and Loss*

The scatterplot between Year and DownTime reveals no evident trend, suggesting that downtime does not change much over the years. But since more downtime values are linked with greater financial losses, the scatterplot between DownTime and Loss points to a positive association. The correlation matrix also demonstrates a modest positive link between DownTime and Loss, further bolstering this relationship. The weak trend of the scatterplot between Year and Loss indicates that the year has little bearing on financial losses directly. All things considered, the matrix plot offers a complete picture of the interactions between the variables, therefore enabling the identification of trends and correlations sometimes not immediately seen from the unprocessed data.

```

Summary Statistics:
> print(summary(matrix_data))
      Year      DownTime      Loss
Min.   :1998   Min.     : 7.00   Min.    :   30
1st Qu.:2002   1st Qu.:19.00   1st Qu.:  726
Median :2009   Median :29.00   Median : 1622
Mean    :2008   Mean    :28.44   Mean    : 2770
3rd Qu.:2013   3rd Qu.:37.00   3rd Qu.: 3083
Max.    :2015   Max.    :50.00   Max.    :18000
>

```

*Figure 33 Summary Statistics of Year, DownTime, and Loss*

Looking at the matrix plot alongside the summary statistics and correlation matrix, we can derive several comprehensive insights about the relationships between Year, DownTime, and Loss in the hacking incidents dataset.

The temporal distribution of incidents, spanning from 1998 to 2015 (17 years), shows an interesting pattern in the data. The mean year of 2008 with a median of 2009 suggests that the dataset is relatively well-balanced across the time period, with a slight skew towards more recent years. This temporal spread allows for robust analysis of how hacking incidents have evolved over time. However, the near-zero correlation coefficient between Year and both DownTime (0.00091) and Loss (-0.00291) indicates that there hasn't been a significant temporal trend in either the duration or financial impact of hacking incidents over this period. This is particularly noteworthy as it suggests that despite technological advancements, the severity and duration of attacks have remained relatively consistent over time.

The DownTime variable reveals crucial operational impacts of hacking incidents. With a mean of 28.44 hours and a median of 29 hours, the central tendency measures are quite close, indicating a somewhat symmetric distribution of incident durations. The range spans from 7 hours to 50 hours, with the interquartile range (IQR) between 19 and 37 hours, suggesting that most incidents are resolved within a day and a half, though some can extend beyond two days. This distribution of downtime provides valuable insights for incident response planning and resource allocation.

Perhaps the most significant finding comes from the Loss variable and its relationship with DownTime. The financial losses range from \$30 to \$18,000, with a mean of \$2,770 and a median of \$1,622. The substantial difference between the mean and median indicates a right-skewed distribution, suggesting that while most incidents result in moderate losses, there are some high-impact events that significantly pull up the average. The correlation coefficient of 0.38099 between

DownTime and Loss represents a moderate positive correlation, which is clearly visible in the scatterplot pattern. This relationship is particularly important as it quantifies how extended downtimes typically translate to larger financial impacts.

The scatterplot matrix visually reinforces these findings through its distributions. The vertical bands in the Year-related plots confirm the consistent distribution of incidents across time. The DownTime versus Loss plot shows a discernible upward trend, though with considerable scatter, indicating that while longer downtimes generally lead to higher losses, other factors likely influence the final financial impact. The transparency in the plotting (set by `rgb(0, 0, 1, 0.3)`) helps reveal density patterns in the data, particularly useful in areas where many points overlap.

### 3.2.2 Analysis 1-2: Average Monthly Financial Loss by Year

```
#-----
# Analysis 2. Heatmap: Average Monthly Financial Loss by Year
#-----
# Calculate average loss for each month-year combination

monthly_yearly_loss <- aggregate(Loss ~ year + month, data = df, FUN = mean)
```

*Figure 34 Calculate average loss*

```
# Create heatmap: x-axis: month, y-axis: year, fill: average loss
p2 <- ggplot(monthly_yearly_loss, aes(x = factor(month), y = factor(year), fill = Loss)) +
  geom_tile() +
  scale_fill_gradient2(low = "white", high = "red", mid = "pink",
    midpoint = mean(monthly_yearly_loss$Loss)) +
  labs(title = "Average Monthly Financial Losses by Year",
    x = "Month",
    y = "Year",
    fill = "Average Loss ($)") +
  theme_minimal()
```

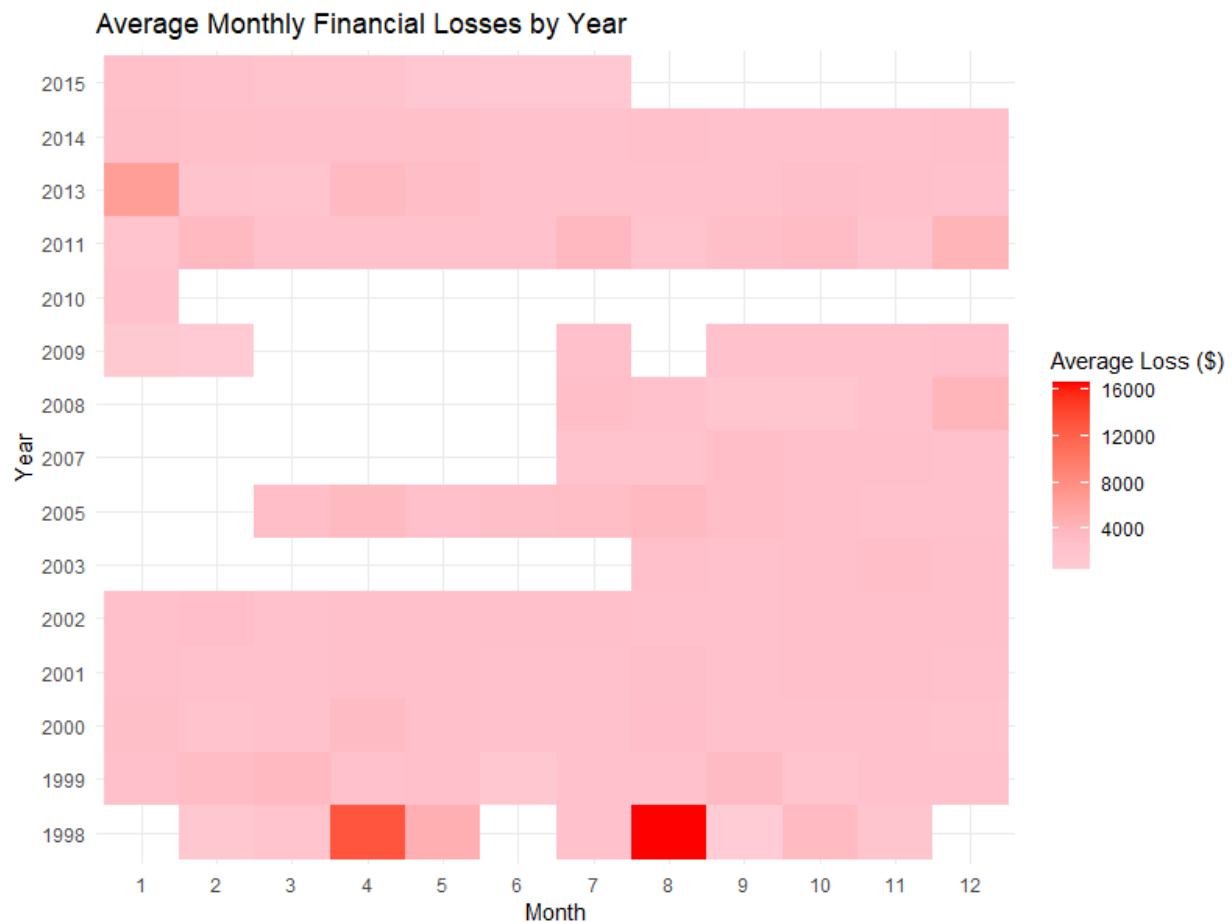
*Figure 35 Generating heatmap*

```
# Save heatmap
png("monthly_yearly_heatmap.png", width = 800, height = 600, res = 100)
print(p2)
dev.off()
```

*Figure 36 Save heatmap*

```
# Print summary statistics
summary_stats <- summary(monthly_yearly_loss$Loss)
print(summary_stats)
```

*Figure 37 Print summary statistics*



*Figure 38 Heatmap of Average Monthly Financial Losses by Year*

The heatmap reveals several important insights about the temporal distribution of financial losses due to hacking incidents. By visualizing the average monthly losses across different years, the heatmap highlights both seasonal patterns and year-to-year variations in the severity of incidents. Certain months, such as those at the end of the year (e.g., October and November), appear to consistently exhibit higher average losses, as indicated by the darker red tiles. This suggests that these months may be associated with increased hacking activity or more severe incidents, potentially due to factors such as heightened business activity or vulnerabilities in organizational

systems during these periods as it's during the holiday season. Conversely, some months, such as January and February, tend to show lighter shades, indicating lower average losses. This seasonal variation could be linked to reduced activity during the start of the year or more effective security measures in place during these months.

The year-to-year trends can provide valuable insights. For example, certain years may indicate a general increase in the intensity of losses throughout numerous months, showing a broader trend of growing financial repercussions from hacking attacks. This could be due to the increased sophistication of cyberattacks or the growing reliance on digital systems, which compounds the potential for financial damage. On the other hand, some years may have relatively lighter hues over most months, indicating periods of lower overall influence. These differences could represent changes in cybersecurity tactics, legislative measures, or even adjustments in the types of attacks being carried out.

```
> summary_stats <- summary(monthly_yearly_loss$Loss)
> print(summary_stats)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   443    2527    2748    2865    2903   16560
> |
```

*Figure 39 Summary Statistics of Average Monthly Losses*

These statistics provide a concise overview of the distribution of average monthly losses across the dataset. The minimum value is 443, indicating that some months experienced very low average losses, while the maximum is 16,560, suggesting that there were months with exceptionally high losses. The first quartile is 2,527 and the median is 2,748, which means that half of the month-year combinations have an average loss below 2,748, and half are above this value. The mean, at 2,865, is slightly higher than the median, indicating a slight right skew in the data, where a few high-loss months pull the mean upward. The third quartile is 2,903, contextualizing that 75% of the values lie below this threshold. Overall, these summary statistics provide insight into the central tendency and variability in average monthly financial losses, which is visually reinforced by the heatmap.

### 3.2.3 Extra Feature

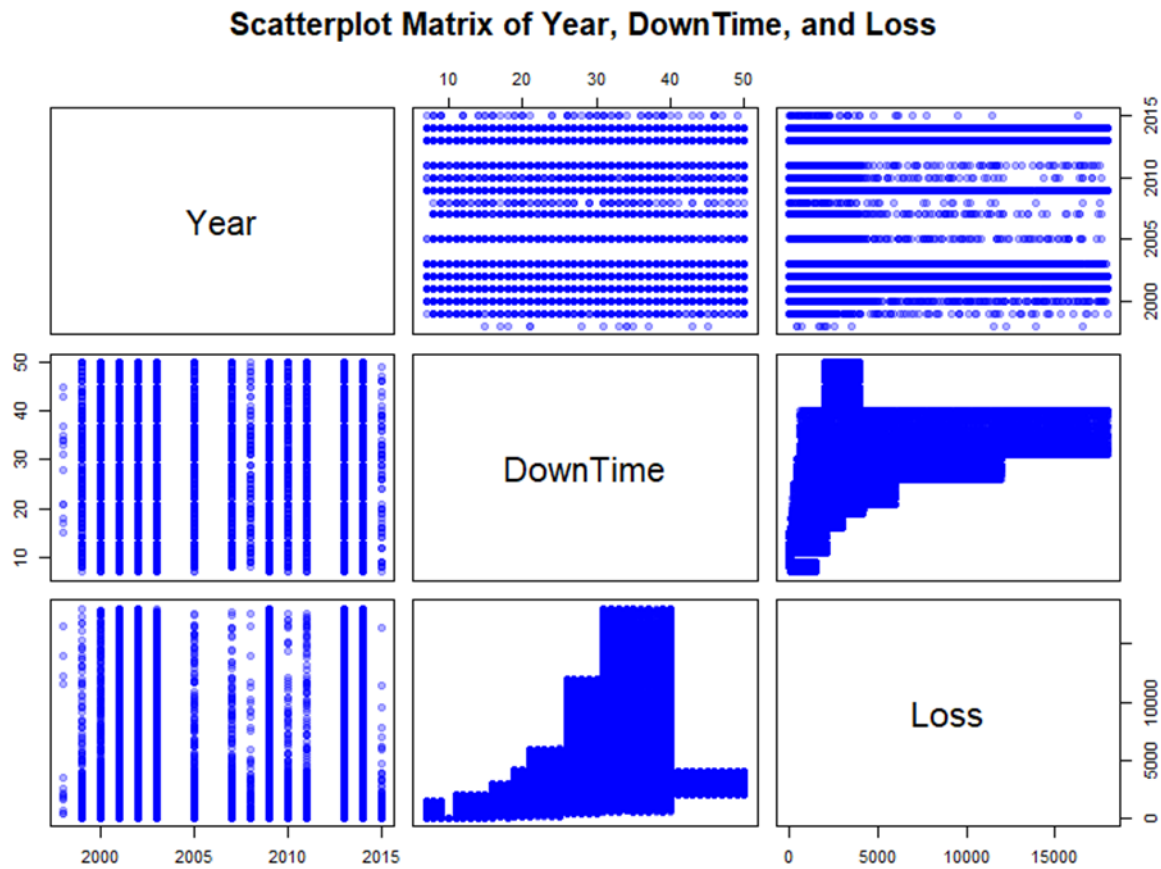
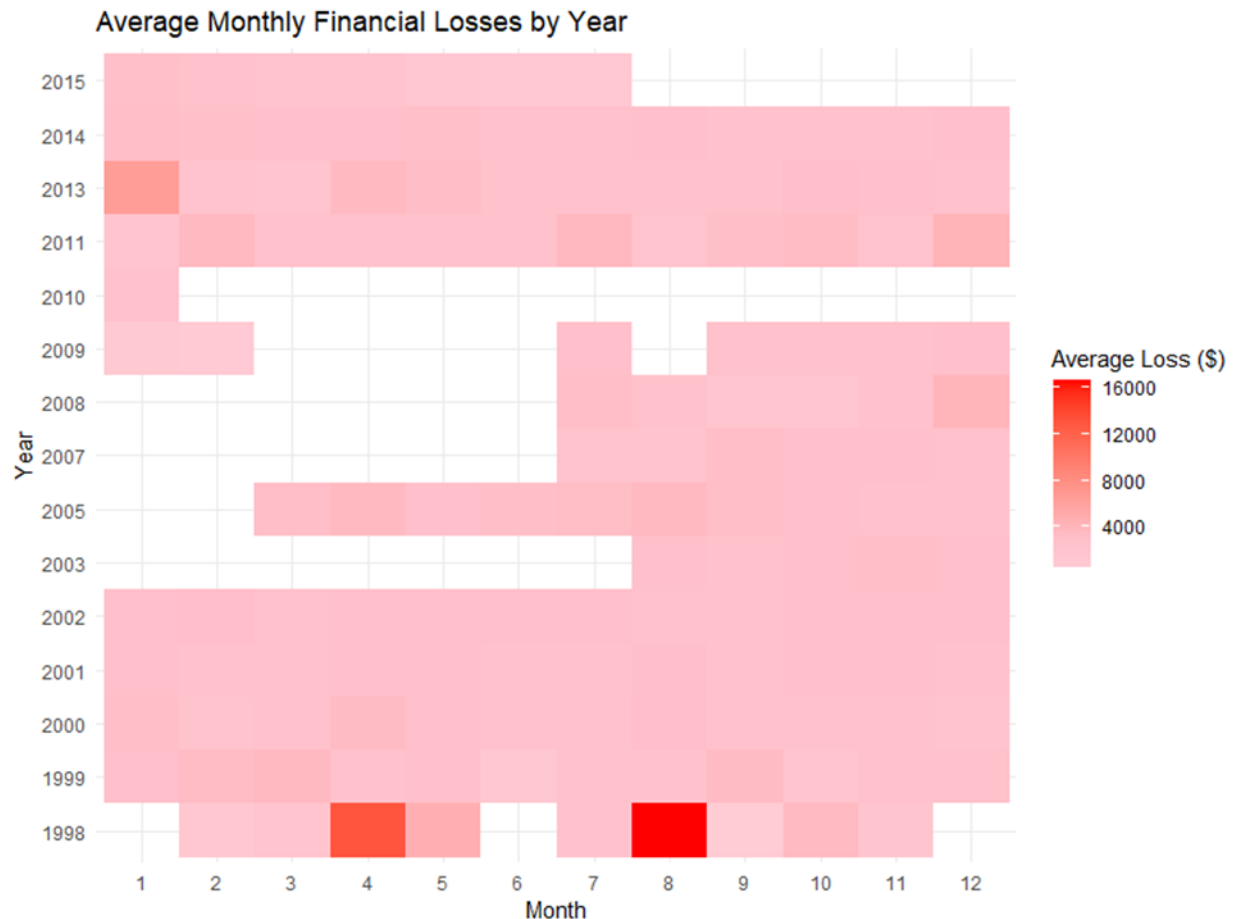


Figure 40 Scatterplot Matrix of Year, DownTime, and Loss



*Figure 41 Heatmap of Average Monthly Financial Losses by Year*

The heatmap and scatterplot matrix offer distinct advantages over traditional plot types for our analysis. The heatmap clearly visualizes average financial losses over time by representing month and year combinations with a color gradient, allowing us to quickly identify periods of high loss. This multi-dimensional display surpasses bar charts or histograms, which typically show single-variable distributions, by exhibiting interactions between time components and loss severity. Meanwhile, the scatterplot matrix efficiently explores pairwise relationships among multiple variables in a compact layout, revealing potential correlations and trends that individual scatter plots or line plots might miss. Both visualizations benefit from the use of colors and shapes that enhance interpretability. They provide a comprehensive overview with less clutter, making it easier to digest complex data patterns and guide further, more focused investigations into factors driving financial losses.

### 3.3 Objective 3: To investigate relationships between web hacking incidents and targeted platforms. – Sailess Raj A/L Rajendran (TP070996)

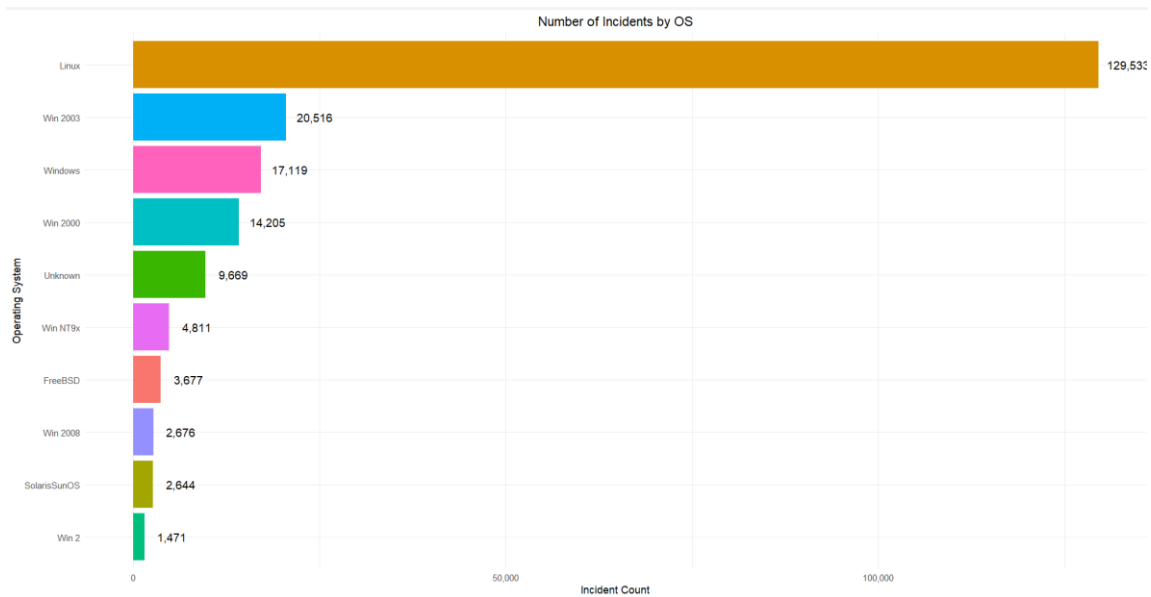
#### 3.3.1 Analysis 1-1: Analysis of the Number of incidents by OS

```
# 1. Number of Incidents by Operating System (OS)
os_counts <- data %>%
  filter(OS != "" & !is.na(OS)) %>%
  group_by(OS) %>%
  summarise(Count = n()) %>%
  filter(Count > 0) %>%
  arrange(desc(Count)) %>%
  head(10) # Show only top 10 OS
```

*Figure 42 shows the codes for the Top 10 Most Affected Operating System by Incident Count*

```
# vertical bar chart for OS incidents
ggplot(os_counts, aes(x = reorder(OS, Count), y = Count, fill = OS)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = format(Count, big.mark=",")), hjust = -0.2, size = 4) +
  coord_flip() +
  theme_minimal() +
  labs(title = "Number of Incidents by OS", x = "Operating System", y = "Incident Count") +
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_y_continuous(labels = scales::comma) |
```

*Figure 43 Vertical Bar Chart Codes for the Top 10 Most Affected Operating System by Incident Count*



*Figure 44 Vertical Bar Chart for the Top 10 Most Affected Operating System by Incident Count*

**Goal:** To analyze how different operating systems are being affected by security incidents, which will recognize the most vulnerable system based on the number of reported incidents.

**Explanation:** This bar chart shows the number of security incidents which is grouped by the operating system. The data has been sorted in a descending order, which shows the top 10 most affected OS platforms. The size of each bar represents the frequency of incidents which has been affected by each of the OS.

- **Interpretation:** The chart shows that Linux encounters the highest figure of security incidents, with 129,533 cases which makes it the most targeted operating system. Windows 2003, Windows and Windows 2000 do show a significant number of incidents, even though much lower than Linux. The presence of incidents in older or less common systems such as Win NT9x, FreeBSD, and Solaris SunOS indicates that vulnerabilities carry on across many types of platforms, even those with a smaller user base. In addition, the **Unknown** category suggests that some incidents which happen on the systems that could not be categorized show up the gaps in system classification and checking.

- Findings:** The overwhelming number of incidents on Linux shows that it is the prime target, which is potentially due to its widespread or most use in server environments. Old versions of Windows, which are Win 2003 and Win 2000, do experience a high attack volume, which is possibly due to outdated security measures. Organizations should give importance in strengthening security measures on these most and frequently targeted OS platforms to mitigate risks.

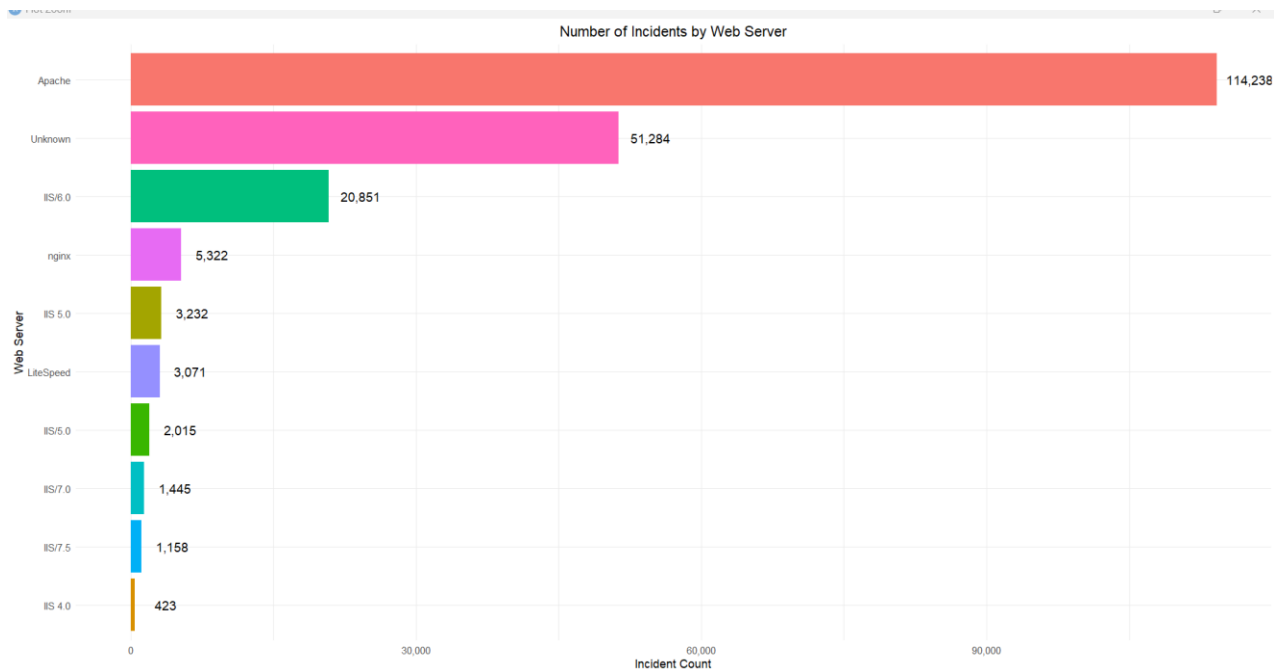
### 3.3.2 Analysis 1-2: Analysis of the Number of incidents by Web Server

```
# 2. Number of Incidents by Web Server Type
webserver_counts <- data %>%
  filter(WebServer != "" & !is.na(WebServer)) %>%
  group_by(WebServer) %>%
  summarise(Count = n()) %>%
  filter(Count > 0) %>%
  arrange(desc(Count)) %>% |
  head(10) # Show only top 10 Web Servers
```

*Figure 45 shows the codes for displaying Top 10 Most Affected Web Servers by Incident Count*

```
#vertical bar chart for web server incidents
ggplot(webserver_counts, aes(x = reorder(WebServer, Count), y = Count, fill = WebServer)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = format(Count, big.mark=",")), hjust = -0.2, size = 4) +
  coord_flip() +
  theme_minimal() +
  labs(title = "Number of Incidents by Web Server", x = "Web Server", y = "Incident Count") +
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5), # Center aligns the title
        axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_y_continuous(labels = scales::comma)
```

*Figure 46 Vertical Bar Chart Codes for the Top 10 Most Affected Web Servers by Incident Count*



*Figure 47 Vertical Bar Chart for the Top 10 Most Affected Web Servers by Incident Count*

**Goal:** The goal of this analysis is to check the distribution of the web server incidents and identify which web servers are most frequently targeted. This will help in getting to know the security risks which will be associated with different web server technologies and the highlights of potential vulnerability in a wide range of used platforms.

**Explanation:** The vertical bar chart shows the number of security incidents which are grouped by the web server type. It shows how different the web servers, which includes the Apache, IIS, nginx and Lite Speed, have been affected by the security threats. The servers are ranked based on the number of incidents, which provides a clear comparison of the security exposure. In addition, an “Unknown” category is included to represent the case where the web server type could not be found.

- **Interpretation:** The Apache web server is the most affected, with 114,238 recorded incidents, significantly higher than any other server type. Whereas the “Unknown” category with 51,284 incidents, which suggest a large number of portion of attacks happens on the unidentified web servers, which indicates a lack of proper monitoring or classification. IIS 6.0, an older Microsoft web server, ranks third with the incidents of 20,851, showing the

outdated system that will stay at a significant security concern. IIS 4.0 and IIS 5.0 suggest that older systems continue to be targeted due to unpatched vulnerabilities and outdated security measures.

- **Findings:** The control of Apache in security incidents suggests that widespread usage makes it a main target for attackers. The high number of incidents in the older IIS versions shows the importance of keeping the software updated, as the timeline of the systems is mainly vulnerable. The huge “Unknown” category will show the potential blind spots in seeing the capabilities, suggesting that the organizations should upgrade the server identification and the logging mechanisms. In conclusion, the need for regular updates, strong security practices, and better conditions in the web server environments to lessen the risk of security breaches.

### 3.4 Objective 4: To Investigate the relationship between key geographical trends and yearly trend comparison of web hacking incidents. – Joshua Emmanuel KC Sathasivam (TP070067)

#### 3.4.1 Analysis 1-1: Geographical Distribution of Web Hacking Incidents

```
# Count Incidents Per Country
country_incidents <- CleanedData %>%
  filter(Country != "UNKNOWN") %>%
  group_by(Country) %>%
  summarise(Incident_Count = n()) %>%
  arrange(desc(Incident_Count))
```

Figure 48 : Code to Count and Rank Incidents Per Country

```
# Display The Top 10 Countries Affected by Web Hacking Incidents
top_10_countries <- head(country_incidents, 10)
```

Figure 49 : Code to Extract The Top 10 Countries Affected by Web Hacking Incidents

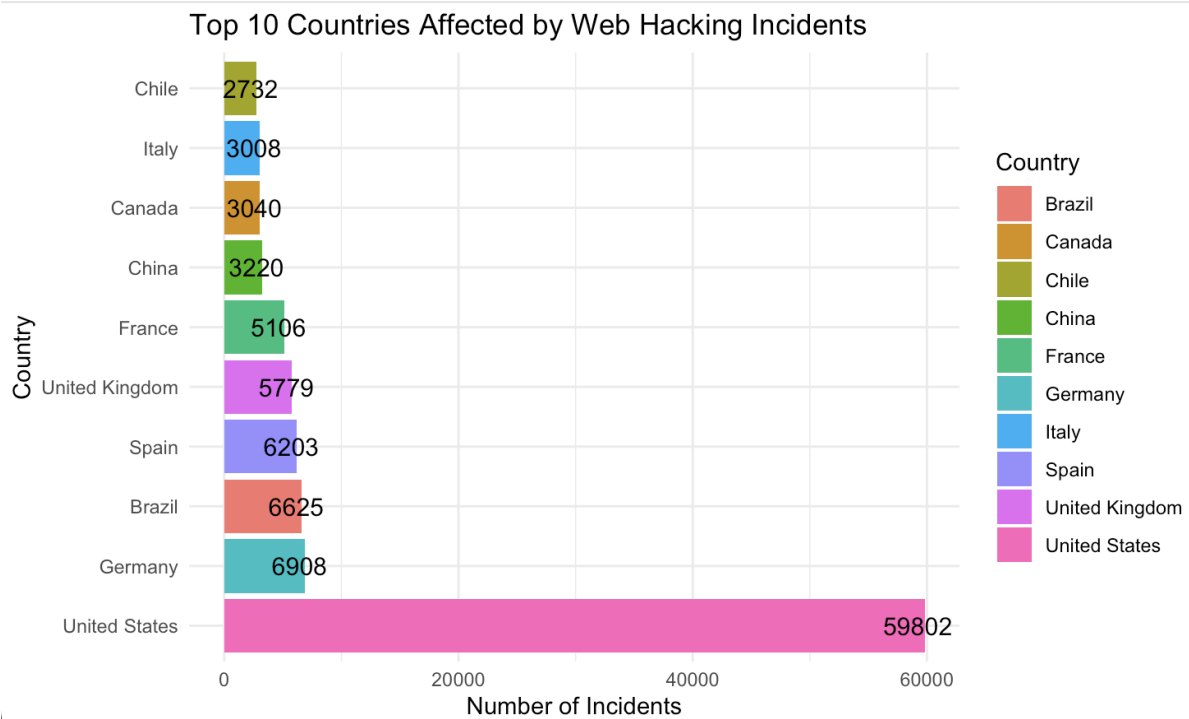
```
# Plot The Top 10 Countries Affected by Web Hacking Incidents
ggplot(country_incidents[1:10, ], aes(x = reorder(Country, -Incident_Count), y = Incident_Count, fill = Country)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Incident_Count), hjust = 0.6, size = 4, color = "black") + # Adding Labels
  coord_flip() +
  labs(title = "Top 10 Countries Affected by Web Hacking Incidents",
       x = "Country",
       y = "Number of Incidents") +
  theme_minimal()
```

Figure 50 : Code to Generate a Horizontal Bar Plot Displaying The Top 10 Countries Affected by Web Hacking Incidents

|    | Country        | Incident_Count |
|----|----------------|----------------|
| 1  | United States  | 59802          |
| 2  | Germany        | 6908           |
| 3  | Brazil         | 6625           |
| 4  | Spain          | 6203           |
| 5  | United Kingdom | 5779           |
| 6  | France         | 5106           |
| 7  | China          | 3220           |
| 8  | Canada         | 3040           |
| 9  | Italy          | 3008           |
| 10 | Chile          | 2732           |

Showing 1 to 10 of 10 entries, 2 total columns

**Figure 51 : Table Displaying The Top 10 Countries Affected and Incident Count of Web Hacking Incidents**



**Figure 52 : Bar Plot Displaying The Top 10 Countries Affected by Web Hacking Incidents**

## Goal

The primary goal of this analysis is to identify and visualize the **geographical distribution of web hacking incidents** by evaluating the number of incidents reported per country. By ranking the most affected countries, the analysis aims to show regions with the highest cyber threats that can be used to implement further cyber security measures.

## Explanation

The analysis above explores the **geographical distribution of web hacking incidents** by utilizing the given source codes, whilst, employing the **dplyr** and **tidyverse** libraries for data manipulation and **ggplot2** library for visualization. The dataset follows a structured approach beginning with **cleaning and filtering out** entries where the country is labelled as "UNKNOWN" to ensure accuracy. The remaining data is then grouped by country, and the number of incidents is counted and arranged in descending order. The top 10 countries with the highest number of incidents are selected and displayed in a table, at the same time, plotted using a horizontal **bar plot**. The analysis above provides an effective method to understand the geographical distribution of web hacking incidents, highlighting countries that are most affected.

- **Interpretation**

The **table** and **bar plot** above shows that the **United States** has the highest number of web hacking incidents, with a count of **59,802**, significantly surpassing other countries. **Germany** follows with **6,908** incidents, while **Brazil (6,625)**, **Spain (6,203)**, and the **United Kingdom (5,779)** also show high number of web hacking incidents. The presence of multiple European nations such as **Germany, Spain, France**, and the **United Kingdom** in the top 10 ranking indicates that the region faces widespread cybersecurity threats. **China, Canada, Italy**, and **Chile** also appear in the top 10, which signifies that cyber threats are not limited to any single region but instead is a global issue. The wide gap between the United States and other countries shows that the United States is a primary target for web hacking, most likely due to its broad digital infrastructure, economic excellence, and various number of internet users.

- **Findings**

The findings also demonstrate several key insights about web hacking incidents. First and foremost, **data standardization** is crucial, as inconsistencies in country names could affect the accuracy of the analysis. Besides that, the **United States** is significantly impacted by web hacking incidents, with considerably higher cases than any other nation. This may be due to its technological advancements, higher digital adoption rates, and the numerous amounts of online businesses. Additionally, European and South American countries also ranked among the most affected, as indicated by the countries such as **Germany, Spain, France, the United Kingdom, and Brazil** in the top 10. Moreover, **China's** inclusion signifies increasing number of web hacking incidents in Asia, which aligns with huge cybersecurity concerns in the region. Last but not least, the analysis above emphasizes the need for **improved cybersecurity measures** to mitigate web hacking incidents across different countries. Addressing data inconsistencies and refining future analysis will improve the efficiency and dependability of insights, allowing for policymakers and cybersecurity experts to develop necessary solutions.

### 3.4.2 Analysis 1-2: Yearly & Geographical Trend Comparison of Web Hacking Incidents

```
# Extract The Year From The Date Column
CleanedData$Year <- year(CleanedData$Date)
```

Figure 53 : Code to Extract The Year From The Date Column

```
# Modify Country Name in The Dataset
top_year_countries_incident <- top_year_countries_incident %>%
  mutate(Country = ifelse(Country == "Netherlands\\", "Netherlands", Country))
top_year_countries_incident <- top_year_countries_incident %>%
  mutate(Country = ifelse(Country == "Southamerica", "South America", Country))
```

Figure 54 : Code to Standardize Country Names in The Dataset

```
# Count Incidents Per Year and Country
yearly_country_trends <- CleanedData %>%
  filter(Country != "UNKNOWN") %>%
  group_by(Year, Country) %>%
  summarise(Incidents = n(), .groups = 'drop')
```

**Figure 55 : Code to Count and Rank The Number of Web Hacking Incidents Per Year and Country**

```
# Display The Most Countries Affected Per Year by Web Hacking Incidents
top_year_countries_incident <- yearly_country_trends %>%
  group_by(Year) %>%
  slice_max(order_by = Incidents, n = 10, with_ties = FALSE)
```

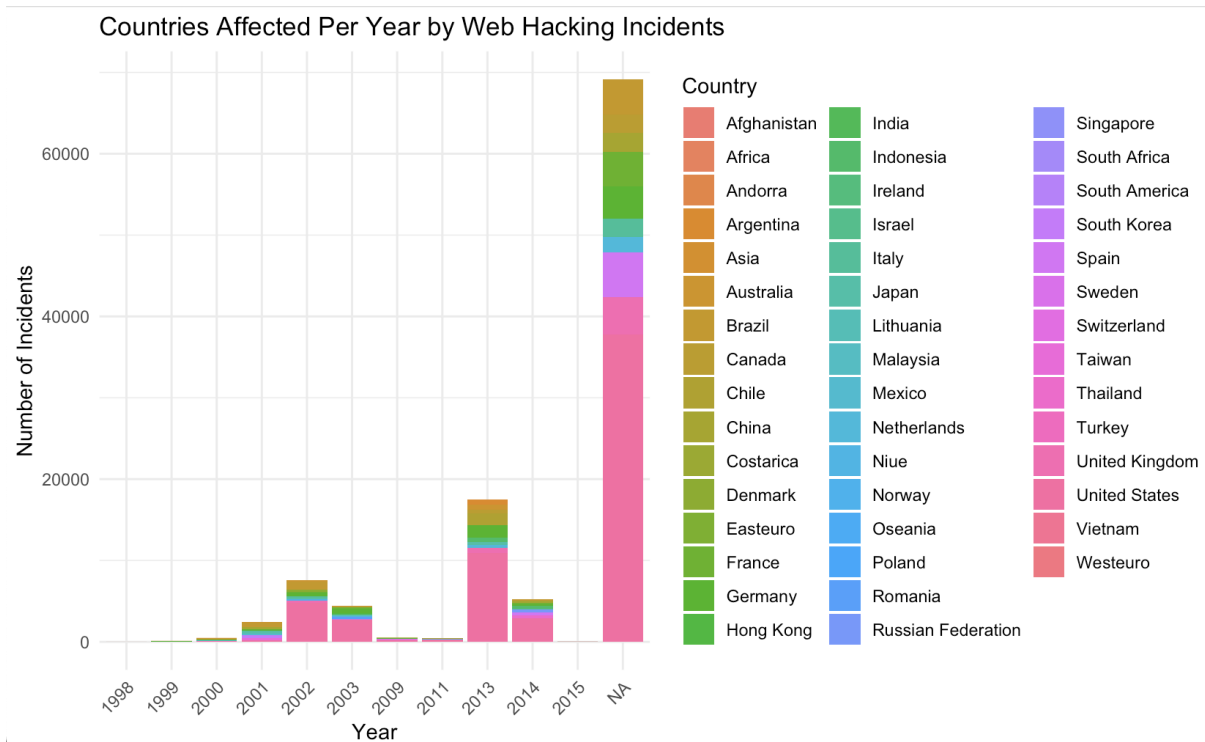
**Figure 56 : Code to Extract The Most Countries Affected Per Year by Web Hacking Incidents**

```
# Plot The Countries Affected Per Year by Web Hacking Incidents
ggplot(top_year_countries_incident, aes(x = factor(Year), y = Incidents, fill = Country)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Countries Affected Per Year by Web Hacking Incidents",
       x = "Year", y = "Number of Incidents") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

**Figure 57 : Code to Generate a Stacked Bar Plot Displaying The Countries Affected Per Year by Web Hacking Incidents**

|                                                 | ▲ Year ▼ | Country ▼      | Incidents ▼ |
|-------------------------------------------------|----------|----------------|-------------|
| 1                                               | 1998     | Afghanistan    | 2           |
| 2                                               | 1998     | Andorra        | 1           |
| 3                                               | 1998     | Costarica      | 1           |
| 4                                               | 1998     | India          | 1           |
| 5                                               | 1998     | United Kingdom | 1           |
| 6                                               | 1998     | United States  | 1           |
| 7                                               | 1999     | Sweden         | 29          |
| 8                                               | 1999     | Germany        | 22          |
| 9                                               | 1999     | Australia      | 7           |
| 10                                              | 1999     | Canada         | 7           |
| 11                                              | 1999     | Brazil         | 6           |
| 12                                              | 1999     | Malaysia       | 6           |
| 13                                              | 1999     | United States  | 4           |
| 14                                              | 1999     | Argentina      | 3           |
| Showing 1 to 14 of 116 entries, 3 total columns |          |                |             |

**Figure 58 : Table Displaying the Year, Countries Affected and Incidents Count of Web Hacking Incidents**



**Figure 59 : Stacked Bar Plot Displaying Countries Affected Per Year by Web Hacking Incidents**

## Goal

The goal of this analysis is to examine both the yearly and geographical trend comparison of web hacking incidents by analyzing the number of incidents reported per country over different years. By identifying yearly trends and the most affected countries each year, the analysis aims to emphasize patterns in cyber threats that can be used to develop further cyber security measures to mitigate threats in years to come and in specific regions.

## Explanation

The analysis above investigates both the **yearly and geographical trends of web hacking incidents** by utilizing the given source codes, whilst, leveraging the **dplyr** and **tidyverse** libraries for data transformation and **ggplot2** library for visualization. The dataset follows a structured approach beginning with **extracting the year from the date column** to facilitate trend analysis. Then, the dataset is **filtered to remove unknown countries**, and the number of incidents is grouped by year and country to track how web hacking incidents have evolved over the years. The dataset is also standardized by modifying inconsistent country names in the dataset to ensure

accurate analysis. The **most countries affected per year by web hacking incidents** is then extracted, and a **stacked bar plot** is generated to visualize the yearly trends of web hacking incidents in different countries. The analysis above provides an effective method to understand the yearly & geographical trend comparison of web hacking incidents.

- **Interpretation**

The **table** and **stacked bar plot** above indicate a rising trend in web hacking incidents over the years. The **early years (1998–2000)** show relatively low web hacking incident counts, but a major spike is observed from **2001 onwards**, with significant rise in **2002, 2013, and 2015**. The **2013 spike** appears significantly unusual, signifying a major cyber threat or a huge increase in web hacking incidents that year. On the contrary, the presence of an "**NA**" year surprisingly stands out with an extremely high web hacking incident count (**over 60,000 incidents**). In terms of geographical trends, the **United States** consistently appears as the most affected country, with a huge amount of web hacking incidents nearly every year in the analysis. European nations like the **United Kingdom, Germany, and France etc.** also appears frequently, indicating widespread web hacking incidents in the region. The **stacked bar plot** also shows a presence of Asian and South American countries such as **Japan, Malaysia, Brazil and Chile etc.** in recent years, signifying that web hacking incidents are becoming more prevalent worldwide rather than being focused in a few regions.

- **Findings**

The findings also demonstrate several key insights about web hacking incidents. First and foremost, **data preprocessing** is necessary to avoid inconsistencies in country names and to maintain the effectiveness of the analysis. In addition, **web hacking incidents** have increased significantly over the years, particularly after **2001**, with major rise in **2002, 2013, and 2015**, indicating emerging cyber threats. On the other hand, the **United States** remains the most frequently affected country by web hacking incidents, increasing the risk of attacks owing to its extensive digital infrastructure and economic accuracy. Concisely, web hacking incidents have grown internationally, with European, South American, and Asian countries such as the **United States** leading, followed by **Germany, Brazil, and China** constantly experience high numbers of web hacking incidents, indicating a transition towards a bigger cyber threat

environment. Last but not least, **data cleaning and validation** is essential for improving the accuracy of the yearly & geographical trend comparison of web hacking incidents.

## **4.0 Conclusion**

### **4.1 Overall Discussion on The Findings From All Objectives**

The thorough study of web hacking trends has given important new perspectives on the frequency, patterns, and effects of web hacking events in several spheres. The first goal exposed a notable rise in web hacking events over the years, especially in times of economic uncertainty and technical development. Important spikes in events between 2002 and 2013 draw attention to how dynamically cyber threats are changing and how new vulnerabilities in web technologies are developing.

The second goal showed a positive link between downtime and financial losses, meaning that extended system failures usually have more financial effects. Still, the steady degree of downtime over the years points to the fact that companies still struggle to shorten the length of delays brought on by cyberattacks, despite technological developments.

Examining the link between web hacking events and targeted platforms, the third aim revealed that Linux and Apache are the most often impacted since they are used in server environments. Emphasising the need of frequent upgrades and patching to reduce risks, the study also highlighted the ongoing threat to antiquated systems including previous versions of Windows and IIS.

With the United States leading in the number of recorded cases, the fourth goal underlined the geographical distribution of cyber hacking events. Its vast digital infrastructure and great internet activity most certainly help to explain this. European and South American countries also saw substantial occurrences, illustrating a global trend in cybersecurity threats. The annual comparison also showed an increase in instances in different areas, therefore highlighting the worldwide reach of web hacking activity.

## 4.2 Recommendation

- 1) **Enhance Monitoring and Reporting:** Encourage broader participation in incident reporting to increase the accuracy and comprehensiveness of threat intelligence.
- 2) **Strengthen Security Measures:** Focus on updating older systems and deploying patches to commonly targeted platforms like Linux and Apache.
- 3) **Incident Response Planning:** Develop ways to reduce downtime and prevent financial damages during cyber events.
- 4) **International Collaboration:** Foster cooperation amongst states to address global cybersecurity concerns and share best practices, exhibiting the rising international breadth of web hacking activity.

## 4.3 Limitation and Future Direction

One disadvantage of our study is the lack of explicit attack methodology in our dataset, which prevents a deeper knowledge of the exact tactics, techniques, and procedures (TTPs) used by hackers. Future study might add more granular data on attack vectors, such as SQL injection, phishing, or malware deployment, to provide more useful information for cybersecurity professionals.

Additionally, our analysis primarily focuses on historical trends, but it does not account for real-time threat intelligence or emerging hacking strategies. Future studies could include live threat feeds and predictive modelling to forecast forthcoming cyber-attacks. Furthermore, increasing the scope to include the impact of emerging technologies, such as artificial intelligence in cybersecurity defence and offense and offer useful perspectives on the future landscape of web security.

**Word Count : 4869**

## 5.0 Workload Matrix

| Component            | Adam Jerome<br>Joseph<br>TP071194 | Hemantharajah<br>A/L<br>Krishnarajah<br>TP070844 | Sailess Raj<br>A/L<br>Rajendran<br>TP070996 | Joshua Emmanuel<br>KC Sathasivam<br>TP070067 | Total |
|----------------------|-----------------------------------|--------------------------------------------------|---------------------------------------------|----------------------------------------------|-------|
| 1.0 Introduction     | 25%                               | 25%                                              | 25%                                         | 25%                                          | 100%  |
| 2.0 Data Preparation | 25%                               | 25%                                              | 25%                                         | 25%                                          | 100%  |
| 3.0 Data Analysis    | 25%                               | 25%                                              | 25%                                         | 25%                                          | 100%  |
| 4.0 Conclusion       | 25%                               | 25%                                              | 25%                                         | 25%                                          | 100%  |

## 6.0 References

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